

Network and Systems Management

Using Game Design Elements in Learning Contexts

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How does GSR work

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Objective

Adopt gaming concepts and elements to improve students' motivation and provide a full set of gameful learning experiences.

How does GSR work

Concepts

Gamification

Using game-thinking and game mechanics to solve problems and to engage audiences.



Gamification

Educational games

- Commercial and wide available games
 - ▶ Civilization
 - ▶ World of Warcraft
- Custom games as learning experiences
 - ▶ Not only a collection of content in a different organization
 - ▶ Should contain the basic game elements
- Games can provide intrinsic motivation to students
 - ▶ Curiosity
 - ▶ Challenges with adaptive difficulty levels
 - ▶ Some degree of chance
 - ▶ An award system
- Can provide more pleasant environment by stimulating creativity

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Award System

Award System

Grading

Assessment and grading follows a reward structure design pattern

- Minimum requirements to succeed
 - ▶ Overcome all the sections, or “levels”
 - ▶ This will grant the 10th grade
- Within each level, the increasing number of overcome obstacles will grant the student with a higher grade
- Soft-currency to stimulate autonomy

Award System

Grading

Element	Description	Grade
Castle	Awarded for each completed level. Student will need five castles to succeed in the course.	Up to 10
Star	Each level will award up to 2 stars.	Up to 10
BitPoint	Awarded when completing a level. Can be used to buy tools or information.	-

Award System

Level map



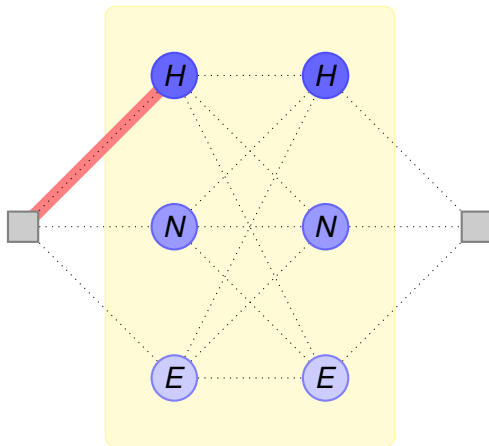
Award System

Stars

Challenge	Description	Stars
Easy	The most simple exercises and games	0
Normal	More complex exercises and games	1
Hard	The most difficult exercises and games	2

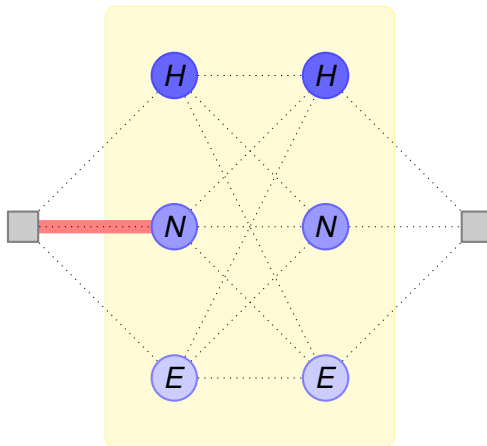
Award System

Grading



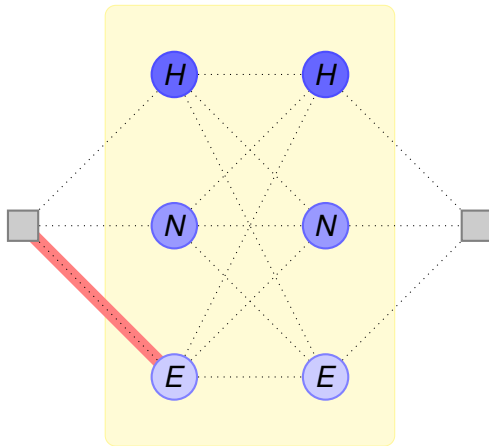
Award System

Grading



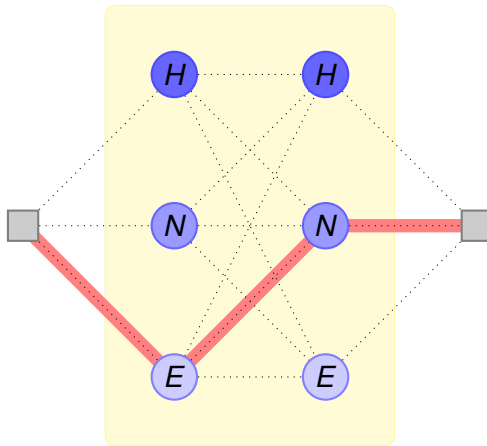
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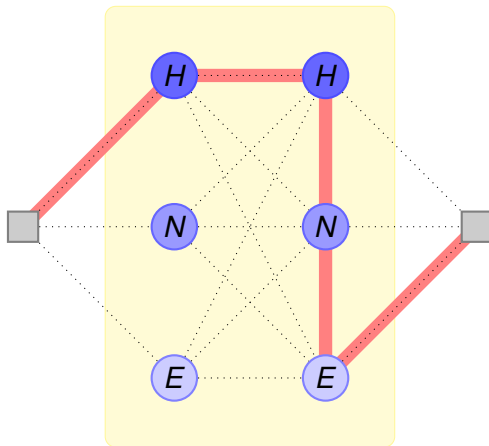
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Grading



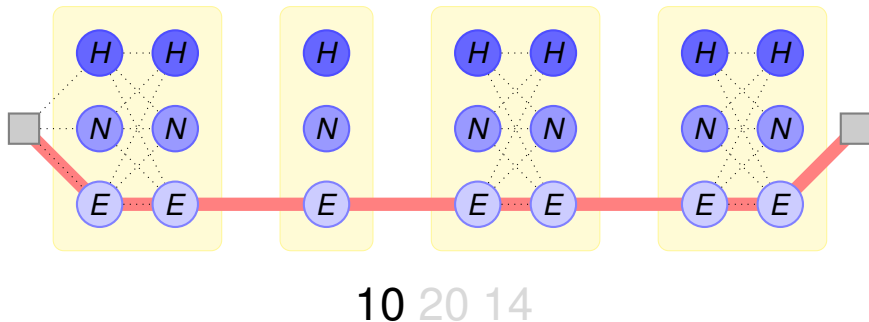
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Grading



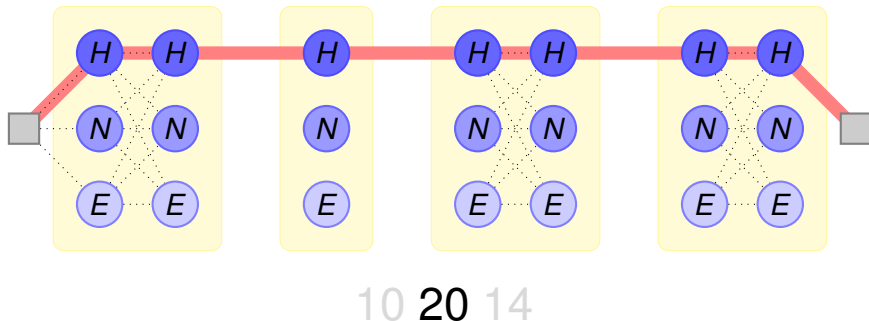
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Grading



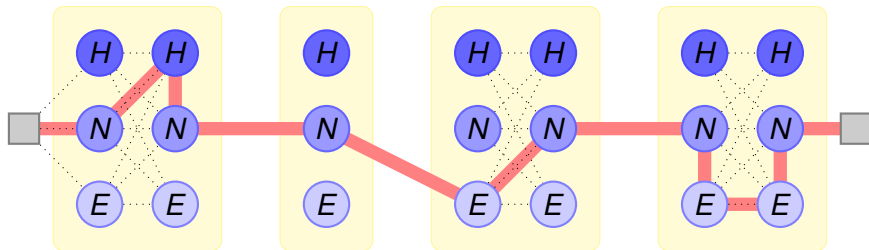
Award System

Grading



Award System

Grading



10 20 14

Award System

Shop



Award System

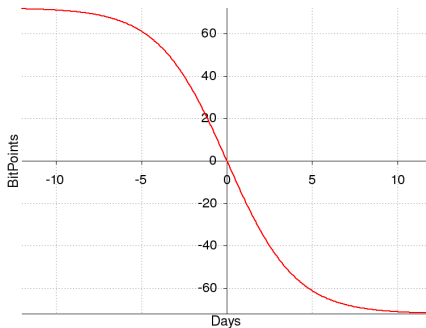
Shop items

Price	Item
50	Command line string
250	Virtual machine file
350	Step-by-step configuration recipe
500	Configuration file
999	Service configured

Award System

BitPoints

- Difficulty
- Creativity
- Time



Learning Experiences

Learning Experiences

Definition

The term Learning Experience describe something the learner is experiencing that hopefully, contributes to a change in thinking, understanding or behaviour.

They should be:

- active
- meaningful
- with social meaning
- integrative, and
- diversified



Learning Experiences

Planification

Level	Title	Aux/Theoretical	Challenges	
1	Virtualization	TO	CG ^E	
			PA ^N	PA ^H
2	Disconnected systems	Slides	PA ^E	
			PA ^N	PA ^H
		Role Play	RPG ^E	
			PA ^N	PA ^H
3	Networked systems	Slides	PA ^E	
			PA ^N	PA ^H
		Slides	PA ^E	
			BG ^N	PA ^H
4	Network Management	Slides	Demo ^E	
			PA ^N	PA ^H
5	Final Integration	TO	PA ^E	
			PA ^N	BG ^H

Learning Experiences

Card Game

The general objective for this game is for students to describe the basic set of virtualization tools.

Learning objectives (at the end, the student should be able to...):

- summarize the main virtualization concepts
- classify virtualization components
- identify advantages and disadvantages of virtualization

The student has to apply a research-develop-use workflow to complete a card game to practice classifying attributes and reinforce factual information about virtualization concepts.

Learning Experiences

Role-Playing Game

The general objective is for students to model the necessary tools to support the persistence of data

Learning objectives (at the end, the student should be able to...):

- point out possible data loss events
- combine tools to overcome harmful situations
- plan approaches using tools and concepts

This is a *tabletop* RPG, meaning that is it played by several students in a social gathering.

Learning Experiences

Board Game – Strategy

The general objective for this game is for students to operate an enterprise wide data-center.

Learning objectives (at the end, the student should be able to...):

- identify all the parts of a networked system
- recognize the dependency between network services and operating systems
- produce a simulation of an enterprise wide data-center
- assess the efficiency of the whole system

Turn based strategy game, where the student will try to manage resources and income.

Conclusions

- Allow the students to choose the final grade
- Provide a set of diversified learning experiences
 - ▶ Practical assignments
 - ▶ Card game
 - ▶ Role-playing game
 - ▶ Board games
- Enjoy learning